Docker Compose stack

Recruiter / Interviewer

Candidate

Web Application · Next.js 14 + TypeScript · port 3000
Recruiter: Home · Policies · Jobs · Candidates · Interviews · Practice · Copilot chat | Candidate: invite link · face lobby · interview room · MCQ quiz
Every screen calls the FastAPI service at /api/v1 (port 8000).

Authentication / Access Control
Recruiter password → JWT (HS256) for recruiter routes · signed candidate and recruiter interview tokens · invite links

Recruiter actions

APPLICATION SERVICES · FASTAPI + WORKERS

Recruitment lifecycle

01 PRE-INTERVIEW

LangGraph: job_creation_graph, cv_screening_graph

1. **Policy upload & index** — Policy PDF or DOCX is split into chunks, embedded with MiniLM and stored in pgvector.
2. **Job creation & post** — Form or copilot chat. JD plus 10 weighted checklist queries from policy excerpts. LangGraph
3. **Receive CVs** — Inbox worker polls Gmail IMAP or Mailpit every 20 s. Recruiters can also upload CVs.
4. **CV processing** — PDF or DOCX up to 8 MB. Native text, then layout order, then Tesseract OCR. CV photo enrolled.
5. **Checklist matching & scoring** — Semantic match, LLM evaluator, Ollama skill recheck, fairness redaction. LangGraph
6. **Ranking & selection** — Met, partial, unmet × weight, plus must-have ratio. Recruiter is notified and picks who to invite.

Telegram channel
Approved job posted by the MCP server (external)

post_job via MCP

Apply mailbox
Gmail IMAP or local Mailpit · CV attachments (external)

CV emails, polled

LiveKit + coturn
WebRTC rooms · TURN relay and public STUN

WebRTC audio and video

Shortlist → invite

02 LIVE INTERVIEW

LiveKit room with the interviewer agent and the transcriber agent

7. **Interview invitation** — Free slot from the schedule rules. Google Calendar when set. Email with interview.ics and a signed link.
8. **Face check & admission** — Candidate consents. Camera frames matched to the CV face with InsightFace. Recruiter admits.
9. **AI video interview** — LiveKit room. Interviewer agent: Whisper listens, Gemini phrases, Kokoro or Piper speaks.
10. **Live transcript** — Transcriber agent joins the room. AssemblyAI speech-to-text. Speaker-labelled lines saved.
11. **In-call tools** — 10-question MCQ from the CV, 10 minutes. Recruiter copilot answers from CV, checklist, transcript.
12. **Call close & handoff** — Session ends. Reports run in order: Q&A, claims, score, quality, assessment.

Report pipeline starts

03 POST-INTERVIEW

LangGraph: qa_analysis_graph, master_assessment_graph · LLM via FallbackLLMManager

13. **Q&A analysis** — Pairs Q&A, then scores relevance, accuracy, completeness and articulation in parallel. LangGraph
14. **CV claim verification** — CV claims checked against transcript quotes. Unresolved claims become follow-up questions.
15. **Interview score** — Six weighted dimensions, technical 24 % down to problem-solving 10 %. CV score is never an input.
16. **Interview quality** — Reviews how the interview was run and recommends changes. Not used for suitability.
17. **Consolidated assessment** — CV 30 % + MCQ 20 % + interview 50 %. Strengths and gaps must cite evidence. LangGraph
18. **Results & reports** — Ranked comparison per role, HTML assessment report, interview packet PDF.

Containers in this layer

frontend · Next.js :3000 | **api** · FastAPI :8000 | **worker** · inbox poll | **interviewer** · LiveKit agent | **transcriber** · LiveKit agent | **mcp** · post_job :8080

SHARED AI SERVICES

LLM / AI processing
FallbackLLMManager: Claude → Gemini → Ollama. Live interviewer line: Gemini only. CV skill recheck: Ollama only.

Speech
faster-whisper (local STT) · Kokoro or Piper (local TTS) · AssemblyAI through LiveKit inference

Face recognition
InsightFace buffalo_l · CV photo enrolment · cosine ≥ 0.42 confirms

Embeddings
all-MiniLM-L6-v2, 384 dimensions · semantic CV-to-checklist match

RAG / policy retrieval
Top policy excerpts feed the job draft and the recruiter copilot

Vector store
Postgres pgvector
policy_chunks.embedding (384)

AI calls

Model providers
Anthropic Claude and Google Gemini (external). Ollama llama3.2 runs locally in Compose.

Invite email + interview.ics

Email invitation
SMTP · invitation email with interview.ics

Free/busy + calendar event

Google Calendar
Free/busy check and event, when configured

Read / write records and evidence

PERSISTENCE · POSTGRES 16 WITH PGVECTOR, PLUS HOST VOLUMES

Async SQLAlchemy 2 · Alembic migrations

Policy & job storage
Policy documents and chunks. Jobs, job versions, checklist queries, postings, scoring progress.
Files: /data/policies

Candidate & CV storage
Candidates, documents, parse results, profiles, CV faces, query evaluations, CV claims, notices.
Files: /data/cvs

Interview & session storage
Interview sessions, magic links, face checks, transcript segments, Q&A pairs, MCQ attempts.
Files: /data/speech models

Scores & assessment storage
Interview analysis, performance, quality. Consolidated assessment, strengths, weaknesses, evidence links, score snapshots.

The locked job version and candidate records stay available to every stage. Recruiters read scores, comparisons and reports through the web application.
Redis 7 is started by Compose and REDIS_URL is set, but no application module opens a Redis client.

Sift
System architecture across the recruitment lifecycle. Numbers follow the demo order.

- Control and data flow
- AI service call
- Live media
- Database access
- LangGraph — Workflow graph